Examiner only

4. An incomplete diagram of the electromagnetic (em) spectrum is shown below.

Region	Radio waves	Microwaves	Infra-red	Visible
Wavelength (m)	....	....	....	....

(a) The wavelength of a wave in each of the regions shown is given below.
Complete the table by using the wavelength values below. [2]

$2 \times 10^{-2}\,m$ $4 \times 10^{2}\,m$ $5 \times 10^{-7}\,m$ $3 \times 10^{-5}\,m$

(b) The ionising regions of the em spectrum are not included on the diagram.

(i) Name the missing regions. [1]

....

(ii) State what is meant by the term ionising radiation. [1]

....

....

(c) All waves in the em spectrum are transverse.
State **one** other property they have in common. [1]

....

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(d) Electromagnetic radiation is used in communications.
A and B are two points on the Earth's surface.
To send a signal from A to B, it can either be sent as a signal in a cable or as a signal via a satellite in geostationary orbit.

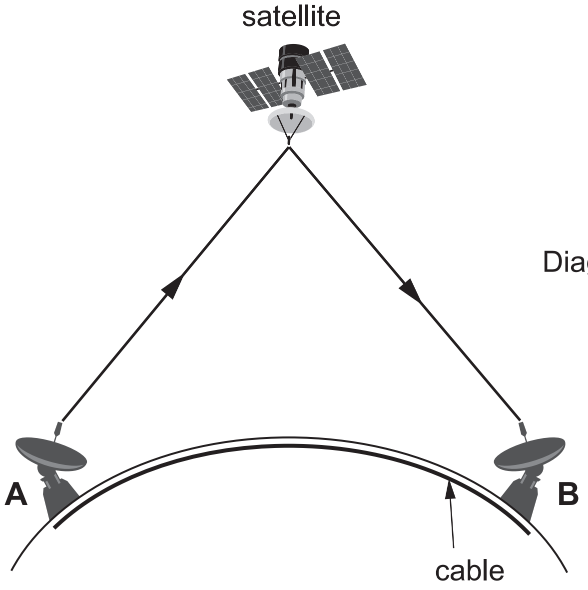

Diagram not drawn to scale

Information about both methods is given in the table below.

Speed of signal to satellite	3×10^8 m/s
Speed of signal in cable	2×10^8 m/s
Distance from A to B through the cable	6×10^6 m
Distance from A to B via the satellite	7.2×10^7 m

Owain states that it is better to send the signal by satellite as it travels faster so it will arrive in a shorter time.
Use data from the table to explain whether you agree. [3]

....

....

8

 (3430UC0-1) **Turn over.**